ROADSENSE | Infrastructure Intelligence

DATE: November 10, 2026
STANDARDS: ASTM D6433 | IRC:SP:77 / IRC:82

ASTM D6433 CONDITION	ROUGHNESS INDEX (IRI)	MATERIAL REQUIRED	ESTIMATED REPAIR BUDGET
10.0 / 100 **Serious / Failed**	**22.43** m/km ASTM E1926	**10520.7 kg** Cold-Mix Bitumen	**Rs. 521164** ($8469.32)

Hazard Classification & Severity Breakdown

Hazard Category	Impact Severity	Count	Recommended Municipal Action
Pothole	Severe (Z >= 3.0)	92	Immediate Cold Patching / Asphalt Mill & Infill (<48h)
Pothole	Moderate (Z 2.0-3.0)	17	Scheduled Maintenance (<14 days)
Pothole	Minor / Anomaly (Z < 2.0)	50	Surface Seal Coating / Monitor
Speedbreaker	Standard Deflection	0	Verify Regulatory Height & Signage Painting

Priority Maintenance Dispatch Table (Top Critical Hazards)

#01	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	0.00000, 0.00000	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay
#02	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	0.00000, 0.00000	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay
#03	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91974, 77.55482	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay
#04	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91974, 77.55482	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay

#05	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91974, 77.55482	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay
#06	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91974, 77.55482	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay
#07	**Pothole** Severe	**Cat Category B (Standard Pothole Cavity)** IRC:SP:77-Clause 4.1	12.91972, 77.55482	3256cm² depth 3.4cm	**13.0 kg**	Rs.886	Square cut, edge priming & Cold-Mix Bituminous Patching
#08	**Pothole** Severe	**Cat Category B (Standard Pothole Cavity)** IRC:SP:77-Clause 4.1	12.91971, 77.55483	3256cm² depth 3.4cm	**13.0 kg**	Rs.886	Square cut, edge priming & Cold-Mix Bituminous Patching
#09	**Pothole** Low	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91970, 77.55483	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay
#10	**Pothole** Low	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91942, 77.55483	13210cm² depth 3.4cm	**52.8 kg**	Rs.2675	Deep milling, granular sub-base recompaction & DBM overlay
#11	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91940, 77.55482	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay
#12	**Pothole** Severe	**Cat Category C (Structural Base / Sub-base Failure)** IRC:82-Clause 6.4	12.91936, 77.55479	25000cm² depth 3.4cm	**99.9 kg**	Rs.4795	Deep milling, granular sub-base recompaction & DBM overlay

Camera Ground-Truth Evidence (AI Detection Bounding Boxes)

Visual verification frames captured by vehicle optical sensors at timestamp of impact.


Pothole: 80%


Pothole: 57%


Pothole: 77%


Pothole: 77%